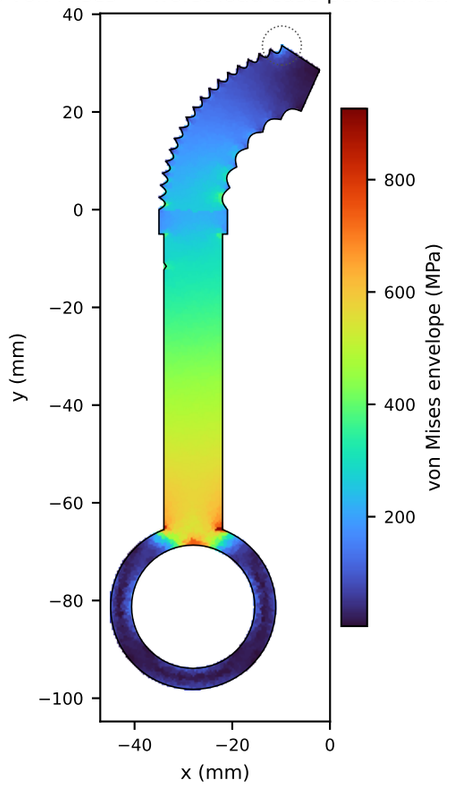
Wave 4 — mh1 merged 10k best (xbabe1_s3005, 17 groups) | Fitness 14.851 N/g | Est. weight 33.4 g

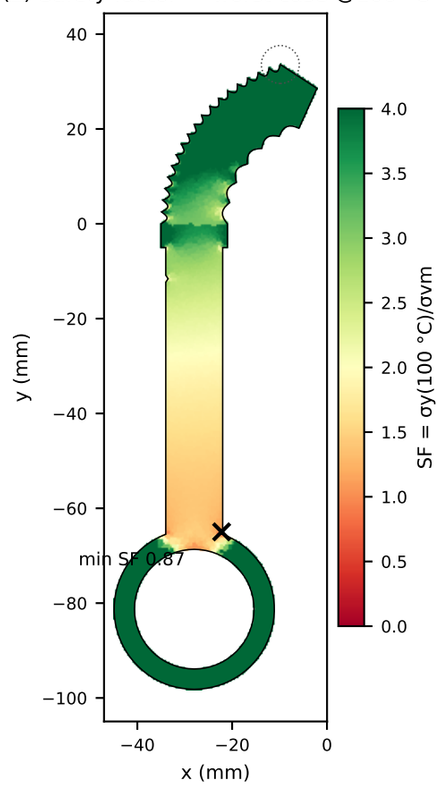
FEM-UNCALIBRATED: offline Kirchhoff plate (numerical DKT + CST, 4715 tris) — not yet calibrated to MC / repo FEM. Envelope of hunter-lateral 260 N, twist 1.26 N·m, tip-pry 260 N, ring-pull 1500 N.

FEM-UNCALIBRATED **QUARANTINED: blade < 3.375 in (all three)** **hard geometric gates: red-team Ok** **floors 180/260/1500 N on plate: FAIL**

(1) von Mises — worst load case per element



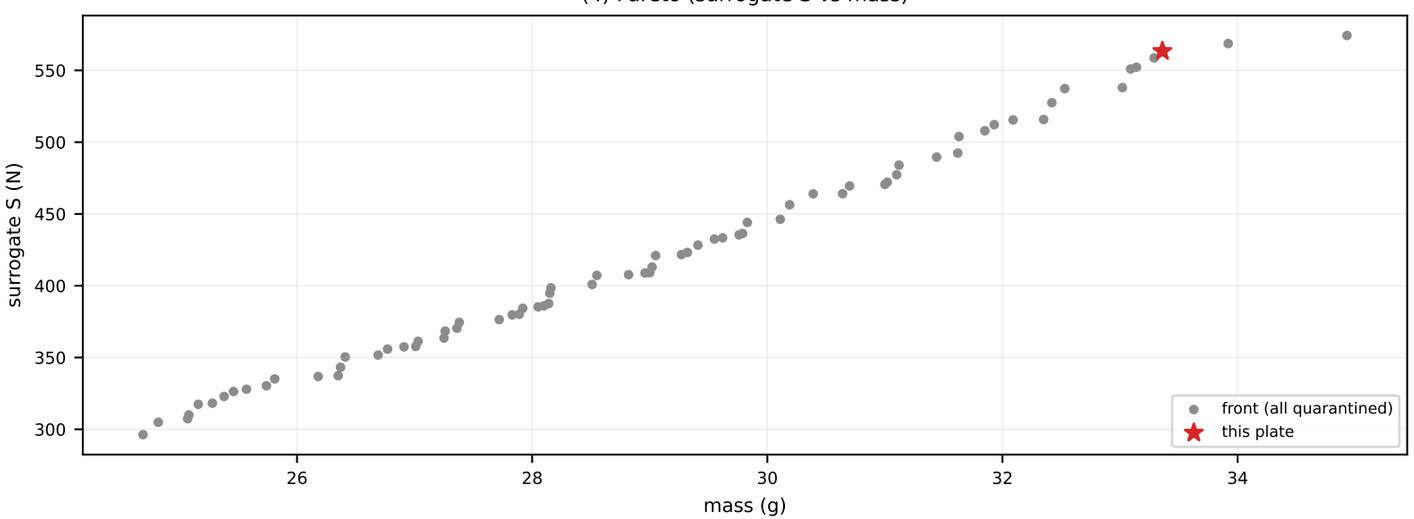
(2) Safety factor — worst case @100 °C



(3) Metrics

Metric	Value	Gate / note
Blade tip→heel	1.715 in (43.5 mm)	**≥3.375 in: FAIL**
Blade horizontal reach	1.294 in (32.9 mm)	**≥3.375 in: FAIL**
Blade edge arc (outer)	2.135 in (54.2 mm)	**≥3.375 in: FAIL**
Grip (ring-neck→guard)	60.5 mm	60–80 mm
Ring ID	25.23 mm	25.4±0.5 mm
Stock / material	4.5 mm / Ti-6Al-4V	red-team: Ok
Est. weight	33.36 g	geom check 33.37 g
Fitness	14.851 N/g	MC p05 strength-per-gram
Beam surrogate S	563 N	repo plate (fem.rs): n/a
Plate FEM @ -10 °C	lat 255 N | ring 5885 N	SF hiker 1.42 / hunter 0.98 / ring 3.92
Plate FEM @ +20 °C	lat 247 N | ring 5697 N	SF hiker 1.37 / hunter 0.95 / ring 3.80
Plate FEM @ +100 °C	lat 225 N | ring 5196 N	SF hiker 1.25 / hunter 0.87 / ring 3.46
Twist / pry SF @100 °C	twist 9.13 | pry 2.55	twist 1.26 N·m, pry 260 N

(4) Pareto (surrogate S vs mass)



Floors on offline plate, worst of -10/20/100 °C: hiker PASS · hunter FAIL · ring PASS
Temp: Ti yield linear derate (-0.11 %/°C from 20 °C); maps use 100 °C yield 803 MPa.
Serration knock-down on blade (h/h_eff=1.21, from history genes). Load patch r≤4 mm excluded from peaks.